The D-QIND Assumption: Decisional Hardness from Preimage Ambiguity and Random Oracles

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Abstract

We introduce the *Decisional Q-Indistinguishability* (D-QIND) assumption, a new standalone hardness assumption in the post-quantum cryptography setting. The Q-Problem framework of Kara et al. defines hardness via indistinguishability over large preimage sets, but its original Q-IND formulation requires the full preimage set $\mathcal{P}_a$ to be explicitly provided to the adversary — a structural requirement that limits applicability. D-QIND removes this requirement: the adversary receives only a single candidate preimage and a hash-based binding tag, and must decide whether the candidate is the true preimage without access to $\mathcal{P}_a$. We prove a hardness lower bound showing that any PPT adversary's advantage against D-QIND is negligible under Pa-Enumeration Hardness and the Random Oracle Model. We establish that D-QIND implies semantic security of the underlying commitment scheme and one-wayness of the binding tag, and we show via a separation argument that D-QIND is orthogonal to — neither implied by nor implying — the original Q-IND assumption. As a proof of applicability, we construct a three-round interactive authentication protocol satisfying Completeness, Soundness, and Honest-Verifier Zero-Knowledge under D-QIND. All constructions are verified concretely via SageMath.

1 Introduction

The advent of large-scale quantum computers threatens the security of virtually all public-key cryptosystems in widespread use today. In response, the post-quantum cryptography (PQC) community has invested heavily in hardness assumptions that resist quantum attacks, most notably Learning With Errors (LWE) [3], the Short Integer Solution (SIS) problem, and code-based and isogeny-based assumptions. Recently, Kara et al. introduced the *Q-Problem* (QP) framework [1], a new paradigm for post-quantum hardness based on the indistinguishability of preimages under a generic data-hiding transformation. The Q-Problem framework is notable for its generality: it does not commit to a specific algebraic structure and subsumes a broad family of candidate hard problems under a unified formalism.

A central notion in the QP framework is *Q-Indistinguishability* (Q-IND), which captures the hardness of identifying a distinguished preimage among a large set of equally valid ones. However, the Q-IND game as formulated by Kara et al. has a structural requirement that limits its applicability: the challenger must explicitly compute and hand the *full* preimage set $\mathcal{P}_a$ to the adversary. This requirement is problematic in two respects. First, it forces the challenger to expose far more information than is necessary for the hardness argument — the adversary receives the entire set $\mathcal{P}_a$ when in practice only a single candidate needs to be evaluated. Second, it ties the game's instantiability to the challenger's ability to enumerate and present $\mathcal{P}_a$, which may be impractical for large or structured preimage sets.

Our Contribution. We address this gap by introducing the *Decisional Q-Indistinguishability* (D-QIND) assumption, a new standalone hardness assumption that requires no enumeration or exposure of $\mathcal{P}_a$. In D-QIND, the adversary receives only a single candidate preimage (x^*, y^*) together with a hash-based binding tag τ, and must decide whether (x^*, y^*) is the true preimage committed in τ or a random valid impostor. This decisional formulation is strictly stronger than Q-IND and, as we show, orthogonal to it: the two assumptions are incomparable under black-box reduction.

Concretely, our contributions are as follows:

- **D-QIND assumption** (Section 3): We define the D-QIND game formally, introduce the auxiliary notion of *Pa-Enumeration Hardness* (PEH), and prove a hardness lower bound showing that any PPT adversary's advantage is bounded by $\mathsf{Adv}^{\mathrm{ROM}} + \mathsf{Adv}^{\mathrm{PEH}} + 1/|\mathcal{P}_a|$, all of which are negligible by assumption.

- **Cryptographic implications** (Section 3): We prove that D-QIND implies semantic security of the hash-based commitment scheme and one-wayness of the binding tag τ.

- **Separation from Q-IND** (Section 3): We show that D-QIND is not black-box reducible to Q-IND, establishing it as a genuinely new member of the post-quantum hardness landscape.

- **Interactive authentication protocol** (Section 4): We construct a three-round Sigma protocol Π and prove Completeness, Soundness, and Honest-Verifier Zero-Knowledge under D-QIND in the Random Oracle Model.

- **Concrete verification** (Appendix A): All constructions are verified via SageMath for the instantiation $f(x, y) = (x + y) \bmod p$ with $p \approx 2^{128}$, confirming $|\mathcal{P}_a| \geq 2^{128}$, $\mathsf{SD}(D_0, D_1) = 1/p$, $I(b\,;\tau) = 0$, and protocol completeness.

Paper Organization. Section 2 recalls necessary preliminaries. Section 3 introduces and analyzes the D-QIND assumption. Section 4 presents the authentication protocol. Section 5 discusses limitations and parameter considerations. Section 6 outlines future work, and Section 7 concludes.

2 Preliminaries

2.1 Notation and Conventions

Let $\lambda \in \mathbb{N}$ denote the security parameter throughout. We write $\mathsf{negl}(\lambda)$ for any function that vanishes faster than the inverse of any polynomial in λ. A probabilistic polynomial-time (PPT) algorithm runs in time poly(λ). We write $x \xleftarrow{\$} X$ to denote sampling x uniformly at random from a finite set X. All cryptographic constructions in this paper are analyzed in the *Random Oracle Model* (ROM), where a hash function $\mathcal{H} : \{0,1\}^* \to \{0,1\}^\lambda$ is modeled as a truly random function accessible to all parties as an oracle. We use $\|$ to denote string concatenation.

2.2 Q-Problem Framework

We briefly recall the Q-Problem (QP) framework introduced by Kara et al. [1], which serves as the conceptual foundation of this work.

A *Q-expression* is a data-hiding transformation $\mathsf{Qe} : \mathcal{D} \times \mathcal{D} \to \mathcal{D}$ of the form $a = x \star y$, where $\mathcal{D}$ is a domain of digital structures and $\star$ is a generic operation over $\mathcal{D}$. The Q-Problem paradigm requires that for a given output a, the set of valid input pairs (x, y) satisfying $x \star y = a$ contains at least 2^I indistinguishable elements, for an indistinguishability parameter I.

The original *Q-Indistinguishability* (Q-IND) assumption [1] is defined via a game in which the adversary receives the full preimage set $\mathcal{P}_a$ explicitly from the challenger and must identify the index j^* of the true preimage. The Q-IND advantage is:

$$\mathsf{Adv}_{\mathcal{A}}^{\text{Q-IND}}(I) \;:=\; \left| \Pr[j' = j^* \mid \mathcal{A}_2(a, \mathcal{P}_a)] - \frac{1}{2^I} \right|.$$

The gap we address. The Q-IND game requires $\mathcal{P}_a$ to be explicitly computable and handed to the adversary by the challenger. This structural requirement limits its applicability in settings where $\mathcal{P}_a$ cannot be efficiently enumerated or should not be exposed. Our work removes this requirement entirely: in D-QIND, the adversary receives only a single candidate preimage and must decide its authenticity without access to $\mathcal{P}_a$.

2.3 Sigma Protocols

A *Sigma protocol* $\Sigma = (\mathcal{P}, \mathcal{V})$ is a three-round interactive proof system between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$ for a relation $\mathcal{R} = \{(x, w)\}$, where x is a public instance and w is a secret witness. The protocol satisfies:

- **Completeness.** For all $(x, w) \in \mathcal{R}$, an honest prover is accepted with probability 1.
- **Special Soundness.** From any two accepting transcripts (a, c, z) and (a, c', z') with $c \neq c'$, a witness w can be efficiently extracted.
- **Honest-Verifier Zero-Knowledge (HVZK).** There exists a PPT simulator $\mathcal{S}$ that, without access to w, produces transcripts identically distributed to those of an honest execution, provided the verifier samples its challenge uniformly.

Protocol Π in Section 4 is constructed as a Sigma protocol where the witness is (s, x, y) and the relation is defined by the D-QIND binding condition.

3 The D-QIND Assumption

3.1 Preimage Sets and Enumeration Hardness

Definition 1 (Preimage Set)**.** *Let $f : \mathcal{D} \times \mathcal{D} \to \mathcal{D}$ be a Q-expression and let $a \in \mathcal{D}$. The* preimage set *of a under f is defined as*

$$\mathcal{P}_a \;=\; \{\, (x, y) \in \mathcal{D} \times \mathcal{D} \mid f(x, y) = a \,\}.$$

We require $|\mathcal{P}_a| \geq 2^\lambda$ for all a in the image of f, where λ is the security parameter.

Definition 2 (Pa-Enumeration Hardness (PEH))**.** *Let f be a Q-expression. We say f satisfies* Pa-Enumeration Hardness *if for every PPT adversary $\mathcal{E}$, for every a in the image of f, and for any secret $s \in \mathcal{S}$ and binding tag τ honestly generated from a uniformly random preimage $(x, y) \xleftarrow{\$} \mathcal{P}_a$, the probability that $\mathcal{E}(1^\lambda, a, \tau, r)$ outputs the true preimage (x, y) is at most $1/|\mathcal{P}_a| + \mathsf{negl}(\lambda)$:*

$$\Pr\Big[\, (x', y') \leftarrow \mathcal{E}(1^\lambda, a, \tau, r) \;:\; (x', y') = (x, y) \,\Big] \;\leq\; \frac{1}{|\mathcal{P}_a|} + \mathsf{negl}(\lambda).$$

Equivalently, no PPT adversary can identify the true preimage among all elements of $\mathcal{P}_a$ with non-negligible advantage over random guessing.

Remark. PEH does not assert that $\mathcal{P}_a$ is incomputable or that generating elements of $\mathcal{P}_a$ is difficult. Any party may freely produce valid preimages by sampling x uniformly and computing y accordingly. PEH asserts only that no PPT adversary can identify *which* element of $\mathcal{P}_a$ is the true preimage committed in τ with probability exceeding $1/|\mathcal{P}_a|$ — the baseline of random guessing. This is precisely the indistinguishability paradigm of the Q-Problem framework [1], where hardness arises from the inability to distinguish the true preimage, not from the difficulty of producing valid ones.

3.2 Hash-Based Commitment

Definition 3 (Binding Tag)**.** *Let $\mathcal{S} = \{0,1\}^\lambda$ be the secret space and let $\mathcal{H} : \{0,1\}^* \to \{0,1\}^\lambda$ be a hash function modeled as a random oracle. For a secret $s \in \mathcal{S}$, a preimage $(x, y) \in \mathcal{P}_a$, output $a \in \mathcal{D}$, and a fresh nonce $r \xleftarrow{\$} \{0,1\}^\lambda$, the* binding tag *is*

$$\tau \;=\; \mathcal{H}(\texttt{"bind"} \,\|\, s \,\|\, x \,\|\, y \,\|\, a \,\|\, r).$$

The nonce r is published alongside τ. The pair (τ, r) constitutes the public binding of the instance.

Hiding. *Since $\mathcal{H}$ is a random oracle and s is uniform over $\mathcal{S}$, the tag τ is computationally indistinguishable from a uniform random string, even given (a, r, x', y') for any $(x', y') \in \mathcal{P}_a$.*

Binding. *A computationally bounded adversary cannot produce two distinct tuples (s, x, y) and (s', x', y') that collide under $\mathcal{H}(\texttt{"bind"} \,\|\, \cdot \,\|\, a \,\|\, r)$ except with probability at most $2^{-\lambda}$.*

Remark (Domain Separation). The prefix `"bind"` *is mandatory and must differ from all other prefixes used by $\mathcal{H}$ in this construction to prevent cross-context collisions.*

3.3 The D-QIND Game

Definition 4 (D-QIND Game)**.** *The* Decisional Q-Indistinguishability *game* $\mathsf{Game}_{\mathcal{A}}^{\text{D-QIND}}(\lambda)$ *between a challenger $\mathcal{C}$ and a PPT adversary $\mathcal{A}$ proceeds as follows.*

Setup.

$$
\begin{aligned}
&s \xleftarrow{\$} \mathcal{S}, \quad (x,y) \xleftarrow{\$} \mathcal{P}_a, \quad r \xleftarrow{\$} \{0,1\}^{\lambda}, \quad b \xleftarrow{\$} \{0,1\},\\
&a = f(x,y), \quad \tau = \mathcal{H}(\texttt{"bind"} \,\|\, s \,\|\, x \,\|\, y \,\|\, a \,\|\, r).
\end{aligned}
$$

Challenge. *$\mathcal{C}$ sends to $\mathcal{A}$:*

$$
\begin{cases}
(a,\, \tau,\, r,\, x,\, y) & \text{if } b = 0 \quad \textit{(real preimage)},\\
(a,\, \tau,\, r,\, x^*,\, y^*) & \text{if } b = 1 \quad \textit{(fake preimage)},
\end{cases}
$$

where $(x^*, y^*) \xleftarrow{\$} \mathcal{P}_a \setminus \{(x,y)\}$.

Output. *$\mathcal{A}$ outputs $b' \in \{0,1\}$. The* D-QIND advantage *of $\mathcal{A}$ is defined as*

$$
\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;=\; \left|\, \Pr[b' = b] - \tfrac{1}{2} \,\right|.
$$

Assumption 1 (D-QIND Assumption)**.** *We say the* D-QIND assumption *holds with respect to $(f, \mathcal{H}, \mathcal{S})$ if for all PPT adversaries $\mathcal{A}$:*

$$
\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\le\; \mathsf{negl}(\lambda).
$$

3.4 Hardness Lower Bound

Theorem 1 (Hardness of D-QIND)**.** *Assume $\mathcal{H}$ is a random oracle and f satisfies Pa-Enumeration Hardness (Definition 2). Then for all PPT adversaries $\mathcal{A}$, there exist PPT adversaries $\mathcal{B}$ and $\mathcal{C}$ such that:*

$$
\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\le\; \mathsf{Adv}_{\mathcal{B}}^{\mathrm{ROM}}(\lambda) \;+\; \mathsf{Adv}_{\mathcal{C}}^{\mathrm{PEH}}(\lambda) \;+\; \frac{1}{|\mathcal{P}_a|}.
$$

Proof Sketch. We partition all possible strategies of $\mathcal{A}$ into three exhaustive and mutually exclusive cases.

Case 1: $\mathcal{A}$ ignores τ. If $\mathcal{A}$ does not use τ in its computation, its view reduces to (a, r, x^*, y^*). Since (x^*, y^*) is a valid element of $\mathcal{P}_a$ regardless of whether $b = 0$ or $b = 1$, and $\mathcal{A}$ has no information about b, its best strategy is a uniform random guess. The only additional advantage comes from enumerating $\mathcal{P}_a$ and checking whether (x^*, y^*) is the "first" element, which succeeds with probability at most $1/|\mathcal{P}_a|$. Hence:

$$\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\leq\; \frac{1}{|\mathcal{P}_a|} \;=\; \mathsf{negl}(\lambda),$$

where the last step follows from $|\mathcal{P}_a| \geq 2^{\lambda}$.

Case 2: $\mathcal{A}$ extracts information from τ. If $\mathcal{A}$ uses τ to gain non-negligible information about (x, y), then $\mathcal{A}$ effectively distinguishes the output of $\mathcal{H}$ from a uniformly random string—contradicting the random oracle assumption. We construct $\mathcal{B}$ that uses $\mathcal{A}$ as a subroutine to win the ROM-distinguishing game. Thus:

$$\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\leq\; \mathsf{Adv}_{\mathcal{B}}^{\mathrm{ROM}}(\lambda).$$

Case 3: $\mathcal{A}$ exploits the structure of f. If $\mathcal{A}$ decides by exploiting structural or statistical properties of f (e.g., a non-uniform distribution over $\mathcal{P}_a$), then $\mathcal{A}$ can enumerate $\mathcal{P}_a$ non-uniformly with non-negligible bias—violating PEH. We construct $\mathcal{C}$ that uses $\mathcal{A}$ to break PEH. Thus:

$$\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\leq\; \mathsf{Adv}_{\mathcal{C}}^{\mathrm{PEH}}(\lambda).$$

Combining. Since Cases 1–3 are exhaustive, a union bound yields:

$$\mathsf{Adv}_{\mathcal{A}}^{\text{D-QIND}}(\lambda) \;\leq\; \mathsf{Adv}_{\mathcal{B}}^{\mathrm{ROM}}(\lambda) + \mathsf{Adv}_{\mathcal{C}}^{\mathrm{PEH}}(\lambda) + \frac{1}{|\mathcal{P}_a|}.$$

All three terms on the right-hand side are negligible by assumption, completing the proof. □

3.5 Implications of D-QIND

Theorem 2 (D-QIND implies Semantic Security)**.** *If the D-QIND assumption holds, then the commitment scheme* $\mathsf{Com}(s; x, y, a, r) = \mathcal{H}(\texttt{"bind"} \,\|\, s \,\|\, x \,\|\, y \,\|\, a \,\|\, r)$ *is semantically secure against preimage identification: no PPT adversary, given* (a, τ, r)*, can output the committed preimage* (x, y) *with non-negligible probability.*

Proof Sketch. Suppose PPT adversary $\mathcal{A}$ breaks semantic security with non-negligible advantage. We construct $\mathcal{B}$ that uses $\mathcal{A}$ to win the D-QIND game. $\mathcal{B}$ forwards (a, τ, r) to $\mathcal{A}$, receives (x', y'), and outputs $b' = 0$ if (x', y') matches the challenge preimage, else $b' = 1$. The misclassification probability when $b = 1$ is at most $1/|\mathcal{P}_a|$, giving:

$$\mathsf{Adv}_{\mathcal{B}}^{\text{D-QIND}}(\lambda) \;\geq\; \mathsf{Adv}_{\mathcal{A}}^{\text{Sem-Sec}}(\lambda) - \frac{1}{|\mathcal{P}_a|} - \mathsf{negl}(\lambda).$$

Since $1/|\mathcal{P}_a|$ is negligible, this contradicts D-QIND. □

Theorem 3 (D-QIND implies One-Wayness of τ)**.** *If the D-QIND assumption holds, then no PPT adversary can recover s from (a, τ, r) with non-negligible probability.*

Proof Sketch. If a PPT adversary $\mathcal{A}$ recovers s from (a, τ, r), it can compute $\mathcal{H}(\texttt{"bind"} \parallel s \parallel x \parallel y \parallel a \parallel r)$ for any candidate $(x, y) \in \mathcal{P}_a$ and check equality with τ. This identifies the true preimage (x, y), directly breaking D-QIND. Contradiction. □

3.6 Separation from Q-IND

Proposition 1 (D-QIND is not directly reducible to Q-IND)**.** *There is no black-box PPT reduction from Q-IND to D-QIND.*

Proof Sketch. The Q-IND game exposes the full preimage set $\mathcal{P}_a$ to the adversary; the D-QIND game does not. Any black-box reduction from Q-IND to D-QIND would require the reduction to simulate, for the D-QIND adversary, a view that includes $\mathcal{P}_a$ explicitly. However, in D-QIND the adversary receives only a single candidate preimage (x^*, y^*) and a binding tag τ — it never sees $\mathcal{P}_a$. Conversely, any attempt to construct $\mathcal{P}_a$ from the D-QIND view requires identifying which element of $\mathcal{P}_a$ is the true preimage, which by PEH succeeds with probability at most $1/|\mathcal{P}_a| + \mathsf{negl}(\lambda)$. Hence no black-box PPT reduction can translate a D-QIND adversary into a Q-IND adversary with non-negligible advantage. Contradiction. □

Corollary 1. *D-QIND is a hardness assumption* orthogonal *to Q-IND: it is neither implied by nor implies Q-IND. The two assumptions differ fundamentally in the information exposed to the adversary—Q-IND exposes $\mathcal{P}_a$ explicitly, while D-QIND withholds it entirely.*

4 Interactive Authentication Protocol

4.1 Setup

Let $\mathsf{pp} = (f, \mathcal{H}, \mathcal{D}, \lambda)$ be the public parameters, where f satisfies PEH and $\mathcal{H}$ is a random oracle. The protocol Π involves two parties:

- **Prover** $\mathcal{P}$: holds secret witness $(s, x, y) \in \mathcal{S} \times \mathcal{D} \times \mathcal{D}$ and public values (a, τ, r) satisfying $f(x, y) = a$ and $\tau = \mathcal{H}(\texttt{"bind"} \,\|\, s \,\|\, x \,\|\, y \,\|\, a \,\|\, r)$.
- **Verifier** $\mathcal{V}$: holds only the public values (a, τ, r).

Goal. $\mathcal{P}$ convinces $\mathcal{V}$ that it knows a valid witness (s, x, y) for (a, τ, r), without revealing any information about s, x, or y.

4.2 Protocol Description

The protocol Π proceeds in three rounds as follows.

Round 1 — Commit $(\mathcal{P} \to \mathcal{V})$.

$$\begin{aligned} &m \xleftarrow{\$} \{0,1\}^\lambda, \\ &\mathsf{com} = \mathcal{H}(\texttt{"com"} \,\|\, m \,\|\, \tau \,\|\, a \,\|\, r). \end{aligned}$$

$\mathcal{P}$ sends com to $\mathcal{V}$.

Round 2 — Challenge $(\mathcal{V} \to \mathcal{P})$.

$$c \xleftarrow{\$} \{0,1\}^\lambda.$$

$\mathcal{V}$ sends c to $\mathcal{P}$.

Round 3 — Response $(\mathcal{P} \to \mathcal{V})$.

$$\pi = m \oplus \mathcal{H}(\texttt{"resp"} \,\|\, c \,\|\, \tau).$$

$\mathcal{P}$ sends π to $\mathcal{V}$.

Verification. $\mathcal{V}$ reconstructs

$$m' = \pi \oplus \mathcal{H}(\texttt{"resp"} \,\|\, c \,\|\, \tau)$$

and accepts if and only if

$$\mathsf{com} \;=\; \mathcal{H}(\texttt{"com"} \,\|\, m' \,\|\, \tau \,\|\, a \,\|\, r).$$

Remark (Domain Separation). The prefixes `"com"` and `"resp"` are distinct from each other and from `"bind"`, ensuring no cross-context collisions in $\mathcal{H}$.

4.3 Completeness

Theorem 4 (Completeness). *For all valid witnesses (s, x, y) and all (a, τ, r) generated honestly, $\mathcal{V}$ accepts with probability 1.*

Proof. An honest $\mathcal{P}$ computes $\pi = m \oplus \mathcal{H}(\texttt{"resp"} \,\|\, c \,\|\, \tau)$. The verifier reconstructs $m' = \pi \oplus \mathcal{H}(\texttt{"resp"} \,\|\, c \,\|\, \tau) = m$ exactly. The check $\mathsf{com} = \mathcal{H}(\texttt{"com"} \,\|\, m \,\|\, \tau \,\|\, a \,\|\, r)$ then holds by construction. □

4.4 Soundness

Theorem 5 (Soundness). *For all PPT cheating provers $\mathcal{P}^*$:*

$$\Pr[\mathcal{V} \textit{ accepts} \mid \mathcal{P}^*] \;\leq\; 2^{-\lambda} + \mathsf{Adv}^{\mathsf{D\text{-}QIND}}_{\mathcal{P}^*}(\lambda).$$

Proof Sketch. $\mathcal{P}^*$ must commit to com before seeing c. To pass verification, $\mathcal{P}^*$ needs m consistent with both com and the challenge c. Since com is fixed before c is known, predicting c succeeds with probability at most $2^{-\lambda}$ in the ROM.

The remaining advantage requires $\mathcal{P}^*$ to produce a valid τ without knowing the true witness (s, x, y). By Theorem 3, producing a valid τ without the witness implies breaking D-QIND. Hence the total soundness error is bounded by $2^{-\lambda} + \mathsf{Adv}^{\mathsf{D\text{-}QIND}}_{\mathcal{P}^*}(\lambda)$, which is negligible under the D-QIND assumption. □

4.5 Honest-Verifier Zero-Knowledge

Theorem 6 (Honest-Verifier Zero-Knowledge). *Protocol Π is honest-verifier zero-knowledge (HVZK). There exists a PPT simulator $\mathcal{S}$, without access to the witness (s, x, y), that produces transcripts (com, c, π) identically distributed to those of an honest execution.*

Proof Sketch. $\mathcal{S}$ operates as follows:

1. Sample $\pi \xleftarrow{\$} \{0,1\}^\lambda$ and $c \xleftarrow{\$} \{0,1\}^\lambda$.
2. Set $m' = \pi \oplus \mathcal{H}(\texttt{"resp"} \,\|\, c \,\|\, \tau)$.
3. Set $\mathsf{com} = \mathcal{H}(\texttt{"com"} \,\|\, m' \,\|\, \tau \,\|\, a \,\|\, r)$.
4. Output (com, c, π).

In the ROM, π and c are uniformly random, and com is determined by them—matching the honest distribution exactly.

Note. This guarantee holds only for an *honest* verifier that samples c uniformly. Extension to a malicious verifier requires additional techniques and is left as future work (Section 6). $\square$

5 Discussion

5.1 On the Standalone Nature of D-QIND

The D-QIND assumption is self-contained: it does not inherit its hardness from Q-IND, LWE, SIS, or any other established assumption. Its security rests on two independent pillars — Pa-Enumeration Hardness of the underlying Q-expression f, and the random oracle assumption on $\mathcal{H}$. Proposition 1 formalizes that no black-box reduction from Q-IND to D-QIND exists, establishing D-QIND as a genuinely new member of the post-quantum hardness landscape rather than a variant of prior work.

5.2 Parameter Considerations

The primary parameter governing security is the size of the preimage set $|\mathcal{P}_a|$. We require $|\mathcal{P}_a| \geq 2^\lambda$, which for $\lambda = 128$ is satisfied by any Q-expression f over a domain of cardinality at least 2^{128}.

On the concrete instantiation. Throughout this paper, we use $f(x, y) = (x + y) \bmod p$ with $p \approx 2^{128}$ as the concrete instantiation for verifying the statistical properties of the D-QIND game. This instantiation is valid and non-trivial in the sense that is relevant to D-QIND: $|\mathcal{P}_a| = p \geq 2^{128}$, and all elements of $\mathcal{P}_a = \{(x, a - x \bmod p) \mid x \in \mathbb{Z}_p\}$ are computationally indistinguishable from one another — any party can generate a fresh element of $\mathcal{P}_a$ by sampling x uniformly, but no party can identify *which* element is the true preimage committed in τ without knowing the secret s. This is precisely the hardness property required by PEH and D-QIND: not that $\mathcal{P}_a$ is difficult to enumerate, but that no party can enumerate $\mathcal{P}_a$ with any *bias toward the true preimage.* The concrete verifications ($\mathsf{SD}(D_0, D_1) = 1/p$, $I(b\,;\tau) = 0$, and protocol completeness) confirm that this instantiation satisfies all required properties.

Alignment with Q-Problem. This interpretation is consistent with the Q-Problem framework of Kara et al. [1], where hardness is defined through *indistinguishability* of preimages rather than through the difficulty of enumerating the preimage set itself. A Q-expression need not have a computationally inaccessible $\mathcal{P}_a$; it suffices that no adversary can identify the distinguished element among all valid preimages. Our instantiation satisfies this condition exactly.

Domain separation. The prefixes `"bind"`, `"com"`, and `"resp"` must remain distinct across all usages of $\mathcal{H}$ to prevent cross-context collisions. Any concrete deployment must enforce this at the implementation level.

5.3 Limitations

We acknowledge three limitations of the current work. First, Protocol Π achieves only honest-verifier zero-knowledge. The stronger property of full zero-knowledge against a malicious verifier is not established here. Second, the protocol is interactive, requiring both parties to be online simultaneously; asynchronous settings are not supported. Third, while PEH is defined as a standalone assumption, we do not provide a formal reduction from PEH to any previously studied hard problem. Its long-term credibility depends on future cryptanalytic scrutiny. We note, however, that PEH as instantiated in this paper — for $f(x, y) = (x + y) \bmod p$ — is consistent with the Q-Problem indistinguishability paradigm of Kara et al., where hardness arises from the inability to identify the true preimage among all valid ones, not from the difficulty of enumerating $\mathcal{P}_a$.

6 Future Work

Several natural directions extend this work.

Non-interactive variant. Applying the Fiat–Shamir transform [5] to Protocol Π in the ROM yields a non-interactive proof of knowledge, enabling deployment in asynchronous settings such as digital signatures and blockchain protocols.

Full zero-knowledge. Extending Protocol Π to achieve full ZK against a malicious verifier requires a commit-and-reveal mechanism for the verifier's challenge, at the cost of an additional round.

KEM construction. D-QIND may serve as the hardness assumption underlying a key encapsulation mechanism, analogous to the KEM construction of Kara et al. [2] but with the stronger decisional guarantee offered by D-QIND.

Richer instantiations of Q-expressions. While the instantiation $f(x, y) = (x + y) \bmod p$ suffices to verify all properties of D-QIND, exploring Q-expressions with additional structural complexity — such as those based on non-linear Boolean functions, multivariate systems, or code-based constructions — may yield instantiations with stronger or more diverse security properties. This is an open direction of independent interest.

Reduction of PEH. Determining whether PEH reduces to, or is implied by, any known hard problem (e.g., LWE, SIS, syndrome decoding) remains an open question, though we note that such a reduction is not required for D-QIND to be a well-defined and useful assumption.

7 Conclusion

We introduced the *Decisional Q-Indistinguishability* (D-QIND) assumption, a new standalone hardness assumption for post-quantum cryptography. Unlike the original Q-IND assumption of Kara et al. [1], D-QIND does not expose the preimage set $\mathcal{P}_a$ to the adversary; instead, it challenges the adversary to decide the authenticity of a single candidate preimage given only a hash-based binding tag. This decisional formulation is strictly stronger and orthogonal to Q-IND, as established by Proposition 1.

We proved a hardness lower bound (Theorem 1) showing that any PPT adversary's advantage against D-QIND is bounded by the sum of its advantages against ROM-distinguishing, Pa-enumeration, and a negligible random-guessing term. We further showed that D-QIND implies semantic security of the underlying commitment scheme (Theorem 2) and one-wayness of the binding tag (Theorem 3).

As a proof of applicability, we constructed a three-round interactive authentication protocol Π and proved it satisfies Completeness (Theorem 4), Soundness (Theorem 5), and Honest-Verifier Zero-Knowledge (Theorem 6) under the D-QIND assumption in the ROM. All constructions were verified concretely via SageMath, confirming that $|\mathcal{P}_a| \geq 2^{128}$, binding tag hiding, and protocol completeness hold for the instantiation $f(x, y) = (x+y) \bmod p$.

We view D-QIND as a foundational primitive whose full potential — in KEM design, signature schemes, and composable protocols — remains to be explored in future work.

A SageMath Verification

All constructions in this paper were verified concretely using SageMath for the instantiation $f(x, y) = (x+y) \bmod p$ with $p = \texttt{next\_prime}(2^{128})$, using a cryptographically secure random number generator (`secrets.randbelow`).

Three verification scripts were developed and executed; their key results are summarized below.

Script 1: Preimage Set Verification (`verif1.sage`)

Verifies V1 and V2 at $\lambda = 128$ using `Integers(p)` from SageMath for proper modular arithmetic.

V1 — Preimage Set Size.

```
p          = 340282366920938463463374607431768211507
|Pa| >= 2^128: True
FINAL CONCLUSION STEP 1: True
```

V2 — All Preimage Elements Valid. Five independently sampled pairs $(x', a - x' \bmod p)$ confirmed $f(x', y') = a$ for all samples:

```
Sample 1 -> f(x',y') == a? True
Sample 2 -> f(x',y') == a? True
Sample 3 -> f(x',y') == a? True
Sample 4 -> f(x',y') == a? True
Sample 5 -> f(x',y') == a? True
```

Script 2: Binding Tag and Protocol Verification (`verif2.sage`)

Verifies V3 and V4 at $\lambda = 128$ using `secrets.randbelow` and `operator.xor` to avoid SageMath preprocessor issues.

V3 — Binding Tag Hiding. Three fake preimages $(x_f, y_f) \in \mathcal{P}_a \setminus \{(x, y)\}$ produced binding tags $\tau_f \neq \tau$:

```
Fake 1 -> Is tau_f == tau? False
Fake 2 -> Is tau_f == tau? False
Fake 3 -> Is tau_f == tau? False
Verification Result V3 (Hiding): True
```

V4 — Protocol Completeness. Full honest execution of Protocol Π:

```
m_original == m_recovered -> True
com == com_check          -> True
Verification Result V4 (Completeness): True
```

Script 3: Information-Theoretic Verification (`verif3.sage`)

Verifies V5 and V6 at $\lambda = 128$ with $N = 100{,}000$ samples. The challenge bit b is sampled uniformly via `secrets.randbelow(2)` per iteration, ensuring genuine randomness in the D-QIND game simulation.

V5 — Mutual Information $I(b\,;\,\tau) = 0$.

```
b=0 count              : 49995   (~50%)
b=1 count              : 50005   (~50%)
I(b ; tau) observed    : 0.00000000 bits
MI = 0?                : True
```

V6 — Pinsker Bound.

```
SD(D0,D1) analytic     : 2.94e-39
Pinsker bound MI <=    : 2.49e-77
MI <= Pinsker bound    : True
D-QIND secure          : True
```

Summary.

Property	Script	Result	Supports
$\lvert\mathcal{P}_a\rvert \geq 2^\lambda$	verif1	True	Definition 1
All $\mathcal{P}_a$ elements valid	verif1	True	Definition 1
τ unique per witness	verif2	True	Definition 3
Protocol completeness	verif2	True	Theorem 4
$I(b\,;\,\tau) = 0$	verif3	True	Theorem 1
$\mathsf{SD}(D_0, D_1) = 1/\lvert\mathcal{P}_a\rvert$	verif3	True	Theorem 1
Pinsker bound satisfied	verif3	True	Theorem 1

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